Informatica

How to Configure SAP BAPI Connector as a Business Service in Cloud Data Integration

© Copyright Informatica LLC 2017, 2026. Informatica, the Informatica logo, and Informatica Cloud Integration are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https:// www.informatica.com/ trademarks.html.

Abstract

You can configure SAP BAPI Connector as a business service within a mapping or a mapping task for SAP BAPI data integration. To configure SAP BAPI Connector as a business service, you must associate an SAP BAPI connection with the business service, and add the required BAPI operation for the business service. This article describes how to configure SAP BAPI Connector as a business service for SAP BAPI data integration.

Supported Versions

- Cloud Data Integration

Table of Contents

Overview 2
Prerequisites. 2
Step 1. Verify if the Required Licences are Available for SAP BAPI Connector. 3
Step 2. Download and Configure the SAP JCo Libraries for BAPI Data Integration. 3
Configuration tasks. 4
Step 1. Create an SAP BAPI connection. 4
Step 2. Configure an SAP BAPI business service. 6
Step 3. Configure a Web Services transformation in a BAPI mapping 8
SAP BAPI and transformation data types. 10

Overview

You can configure SAP BAPI Connector as a business service within a mapping or a mapping task for SAP BAPI data integration. When you configure SAP BAPI Connector as a business service, you associate an SAP BAPI connection and add the required BAPI operation for the business service. You then create a mapping with a Web Services transformation and associate it with the business service.

When you run the mapping, SAP BAPI Connector constructs an SAP BAPI request based on the source data that you pass, sends the request to SAP, and gets a corresponding response from SAP before you write to the target. In the Web Services transformation, you select elements that you want from the response structure to write to the configured target.

Note: Even though an SAP BAPI mapping runs successfully, the system log of the SAP system displays the CPIC error. To avoid the CPIC error at the SAP system, see the CIPIC error Knowledge Base article.

Prerequisites

Before you can use an SAP BAPI connection for BAPI data integration, you must perform the following prerequisite tasks:

1. Verify if the required licences are available for SAP BAPI Connector.
2. Download and configure the SAP JCo libraries for BAPI data integration.

Step 1. Verify if the Required Licences are Available for SAP BAPI Connector

Before you create an SAP BAPI connection for BAPI data integration, you must verify that the required licences for SAP BAPI Connector are available.

1. In the Data Integration home page, click the **Administrator** tab.
2. Under **Licences**, verify if the **SAP BAPI Connector** licence is enabled.

Step 2. Download and Configure the SAP JCo Libraries for BAPI Data Integration

Before you can use an SAP BAPI connection for BAPI data integration, you must download the SAP JCo libraries from the SAP Service Marketplace and configure them on the machine where the Secure Agent runs.

Contact SAP Customer Support if you encounter any issues with downloading the libraries.

1. Go to the SAP Service Marketplace: http://service.sap.com/connectors

   Note: You will need SAP credentials to access the Service Marketplace.
2. Download the most recent version of the 64-bit SAP JCo libraries based on the operating system on which the Secure Agent runs:

Secure Agent System	SAP File Name
Windows	`sapjco3.jar` `sapjco3.dll`
Linux	`sapjco3.jar` `libsapjco3.so`

3. Copy the JCo libraries to the following directory:

   `<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\ext\deploy_to_main\bin\rdtm-extra\tpl\sap`
4. Configure the JAVA_LIBS property in Data Integration.
   1. Log in to Data Integration.
   2. Click **Runtime Environments** to access the **Runtime Environments** page.
   3. To the left of the agent name, click **Edit Secure Agent**.
   4. From the **Service** list, select **Data Integration Server**.
   5. From the **Type** list, select **Tomcat JRE**.
   6. Enter the JAVA_LIBS value based on the operating system on which the Secure Agent runs.

Operating System	Value
Windows	../bin/rdtm-extra/tpl/sap/sapjco3.jar;../bin/rdtm/javalib/sap/sap-adapter-common.jar
Linux	../bin/rdtm-extra/tpl/sap/sapjco3.jar:../bin/rdtm/javalib/sap/sap-adapter-common.jar

   7. Click **OK** to save the changes.

8. Repeat steps 2 through 7 on every machine where you installed the Secure Agent.

5. Restart the Secure Agent.

Configuration tasks

After you complete the prerequisites, perform the following configuration tasks for BAPI data integration:

1. Create an SAP BAPI connection.
2. Configure an SAP BAPI business service.
3. Configure a Web Services transformation in a BAPI mapping.

   Note: You cannot use the SAP BAPI connection in the Source and Target transformation.

Step 1. Create an SAP BAPI connection

Create an SAP BAPI connection to make a BAPI/RFC call to the SAP system for BAPI data integration. To create an SAP BAPI connection, select the **SAP BAPI** connection type. You can use the SAP BAPI connections in mappings and mapping tasks.

1. Click **Administrator** > **Connections**, and then click **New Connection** to create a connection.

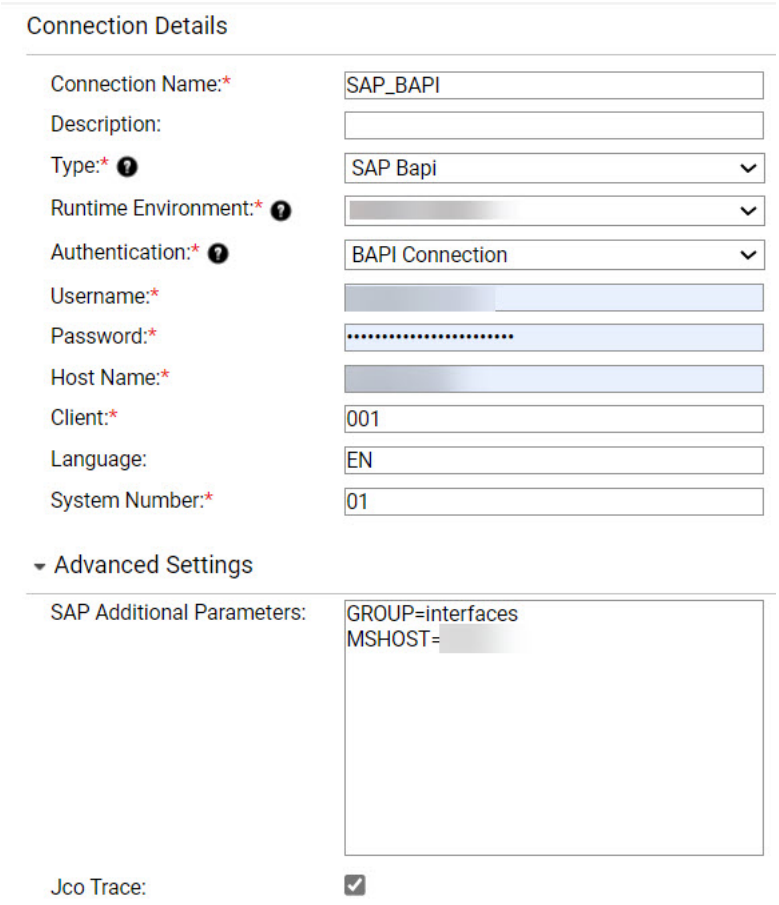
   The **New Connection** page appears.
2. Enter a name for the SAP BAPI connection.

   Connection names are not case sensitive. Connection names can contain alphanumeric characters, spaces, and the following special characters:

   _ . + -
3. Enter a description for the connection.

   The description can have a maximum length of 255 characters.

4. Select **SAP BAPI** as the connection type.



5. Select the name of the runtime environment where you want to run the tasks.
6. Select **BAPI Connection** as the authentication method.
7. Enter an SAP user name with the appropriate user authorization.
8. Enter the SAP password.
9. Enter the host name or IP address of the SAP system that you want to connect to.
10. Enter the SAP client number.
11. Enter the language code that corresponds to the language used in the SAP system.
12. Enter the SAP system number.
13. If you want to enable tracing for the BAPI/RFC call, select the **Jco Trace** check box.

    Note: Even if you define the Trace parameter in the **SAP Additional Parameters** field, you must select the **Jco Trace** check box to generate the trace file.
14. Specify the additional SAP parameters that the Secure Agent uses to connect to the SAP system as an RFC client.

    For example, you can define the following load balancing parameters. SAP identifies the connection type as load balancing based on the defined GROUP and MSHOST parameters and uses those attributes for load balancing.

    ```
    GROUP=interfaces
    MSHOST=<Message server hostname>
    ```

Here, the GROUP parameter defines the group name of the SAP Application Server and MSHOST defines the hostname of the SAP Message Server. For more information about SAP parameters that you can specify, see the SAP documentation.

Note: If you configure a parameter in both the **SAP Additional Parameters** field and the other fields in the connection, the values in the **SAP Additional Parameters** field override the values specified in the connection property fields.

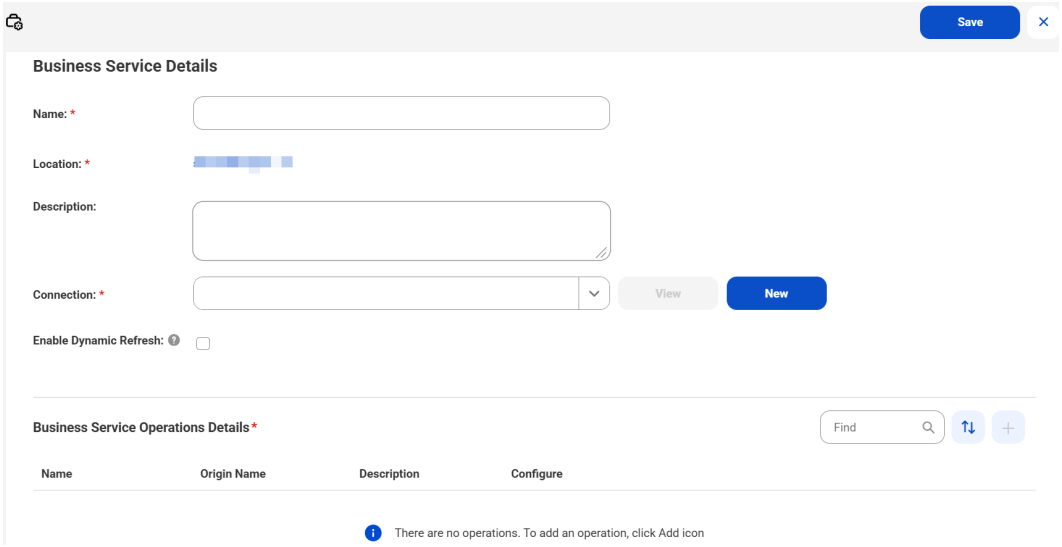
15. Optionally, to commit data to the SAP system with each BAPI/RFC call, enter `DOCOMMIT=true` in the **SAP Additional Parameters** field.
16. Click **Test Connection** to determine if the connection to the SAP system is successful.
17. Click **Save** to save the connection.

Step 2. Configure an SAP BAPI business service

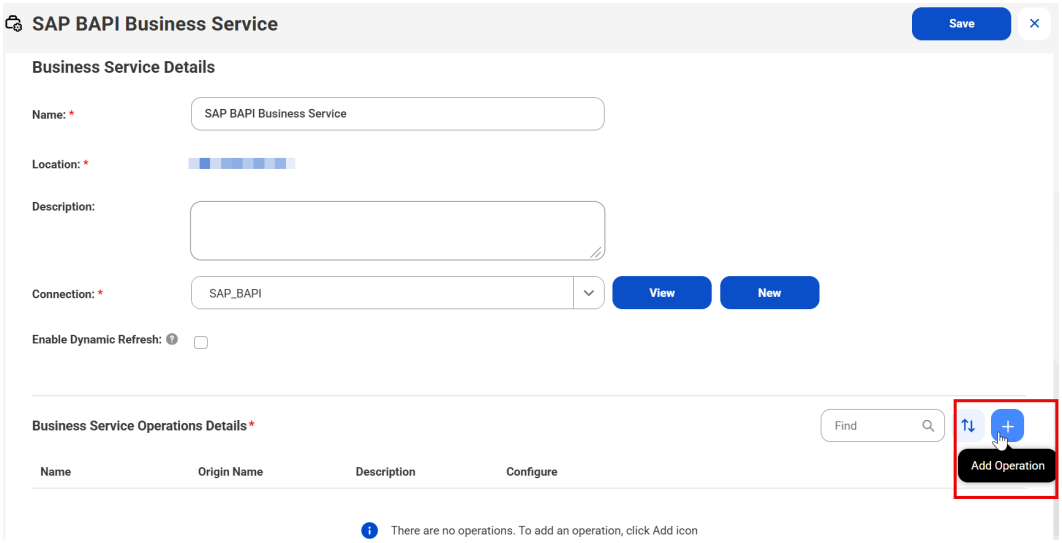
To configure an SAP BAPI business service, select the SAP BAPI connection and the required BAPI operation for the business service.

1. Click **Data Integration** > **New** > **Components** > **Business Services**, and then click **Create**.

   The **Business Service Details** page appears.

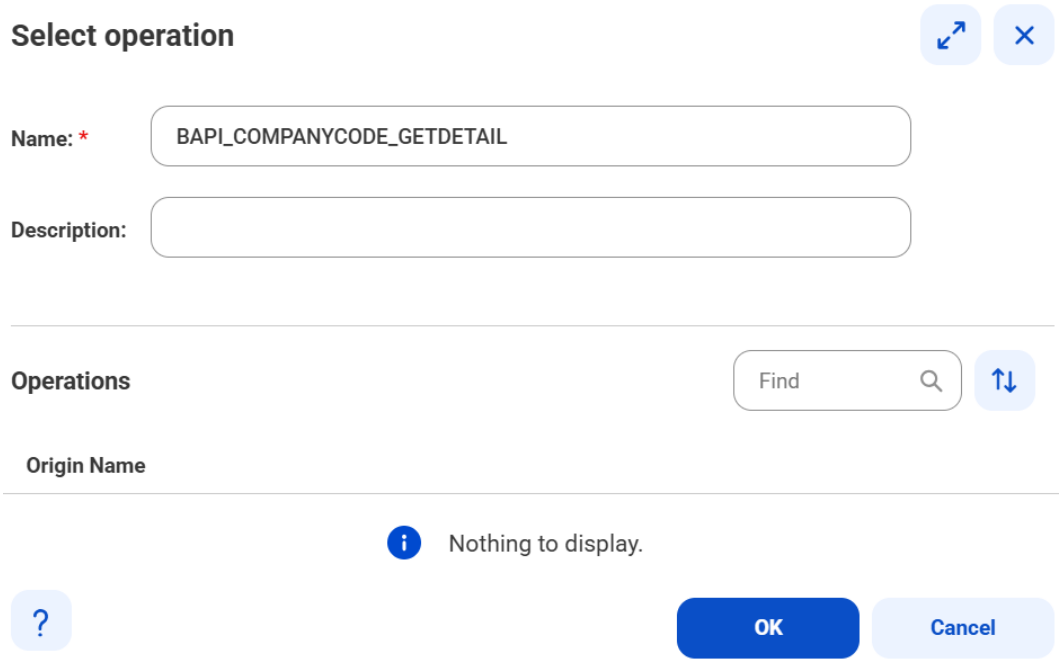


2. Enter a name and description for the business service.
3. Select the SAP BAPI connection that you want to use to make a BAPI/RFC call to the SAP system.
4. Click **Add Operation**.



5. On the **Select Operation** window, enter the full name of the SAP BAPI operation.

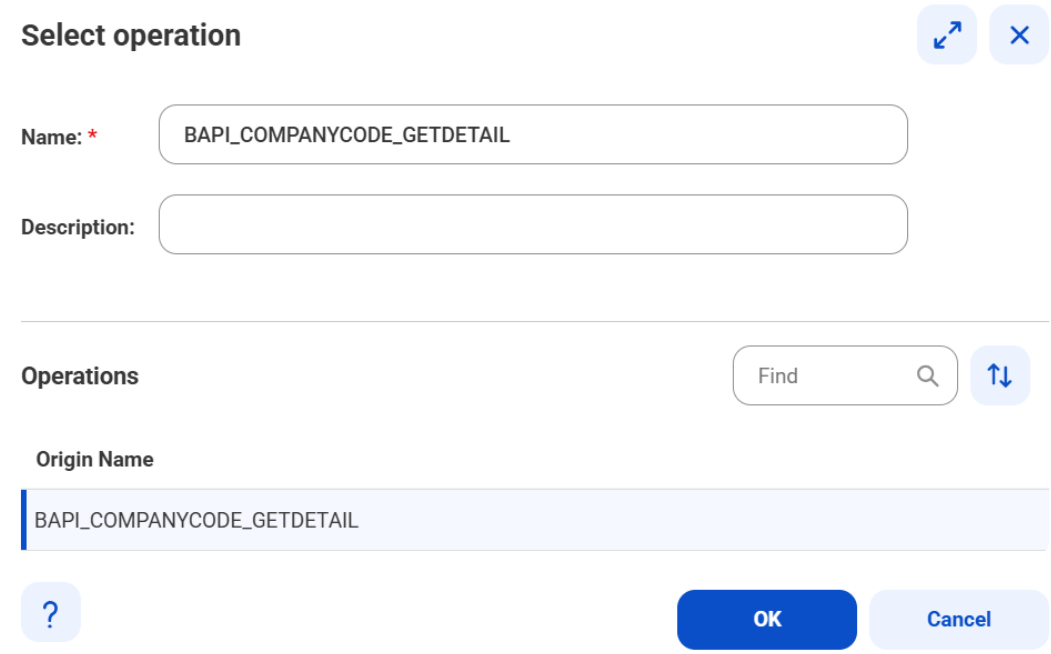
   For example, enter `BAPI_COMPANYCODE_GETDETAIL`.



   Note: If you don't enter the full name of the SAP BAPI operation, you can't fetch the operation.

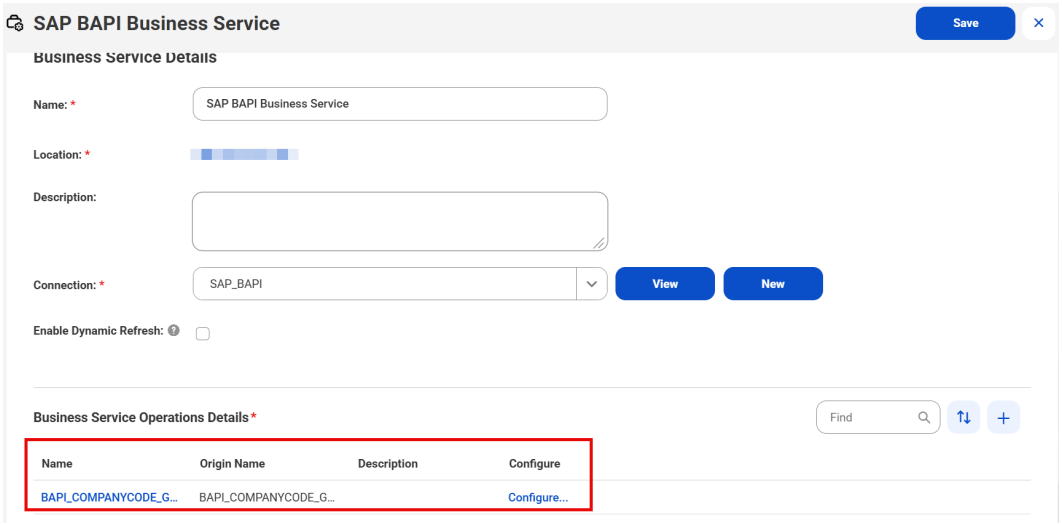
6. To fetch the entered SAP BAPI opertaion, click outside the **Name** field or press `Tab`.

The `BAPI_COMPANYCODE_GETDETAIL` operation fetches as the origin name.



7. In the **Operations** section, select the origin name and click **OK**.

   Repeat steps 4 through 7 to add multiple SAP BAPI operations.



8. Click **Configure** to view the metadata and structure of the SAP BAPI.

   The **Request** tab shows the import details, the **Response** tab shows the export details, and the **Fault** tab shows the fault code details.

Step 3. Configure a Web Services transformation in a BAPI mapping

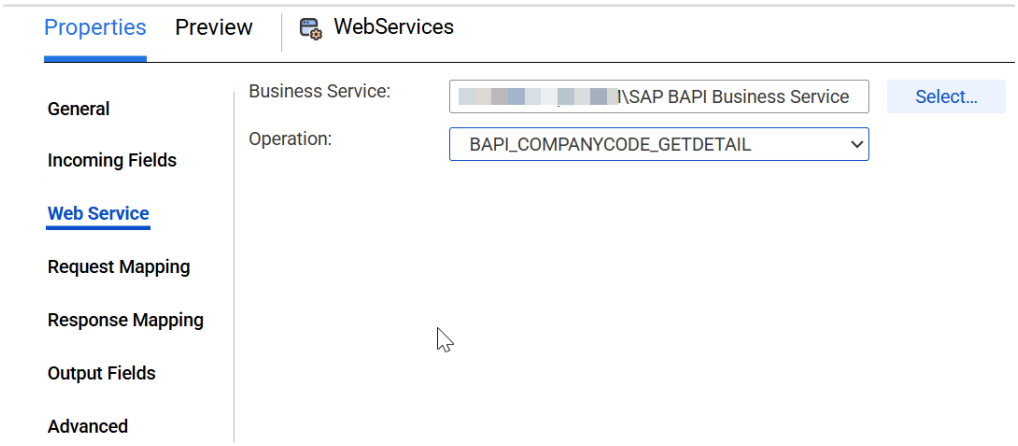
Use the Data Integration Mapping Designer to configure a mapping. Create a Web Services transformation to integrate with SAP BAPIs.

1. To create a mapping, click **Data Integration** > **New** > **Mappings**. Select **Mapping** and click **Create**.

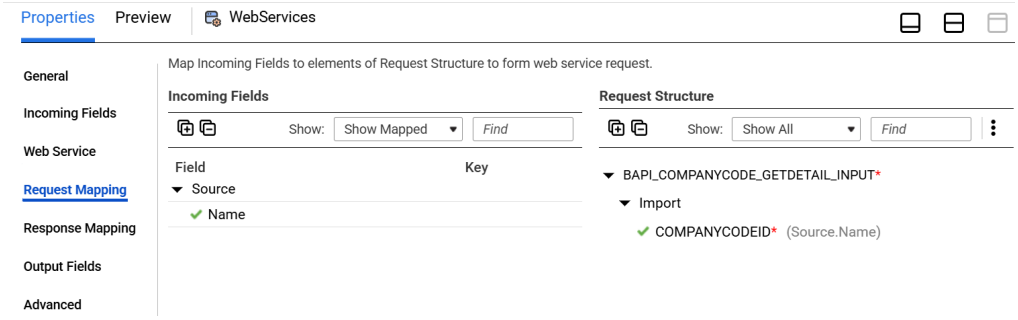
2. Enter a name and a description for the mapping, and click **OK**.

   You can use alphanumeric characters and underscores (_) in the mapping name.

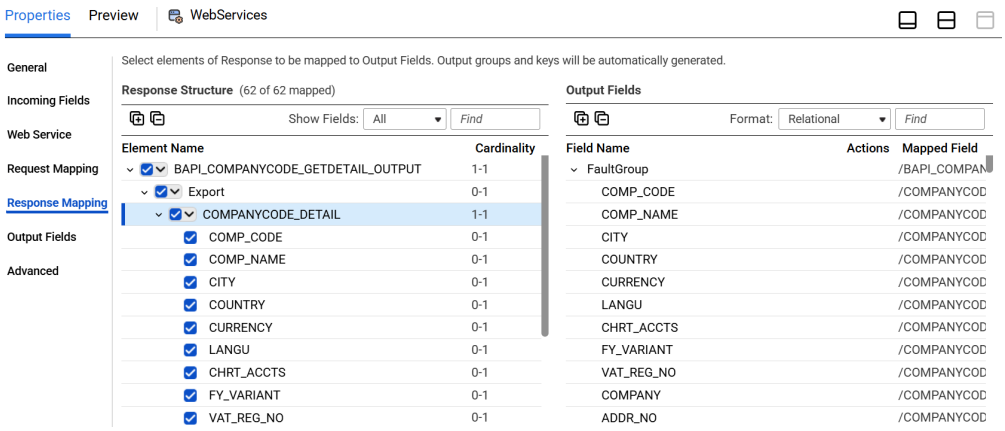
3. To add a Source transformation, on the **Transformation palette**, click **Source**.
   a. On the **General** tab, enter a name and description for the source.
   b. On the **Source** tab, specify a connection.
   c. Add a source object to provide inputs to the Web Services transformation.
   d. In the **Advanced** section, set the tracing level to normal.
4. To add a Web Services transformation, on the **Transformation palette**, click **Web Services**.
   a. On the **General** tab, you can enter a name and description for the Web Services transformation.
   b. Click the **Web Service** tab, select the configured SAP BAPI business service, and then select the required SAP BAPI operation.



   c. Draw a link to connect the Source transformation to the Web Services transformation.
   d. On the **Request Mapping** tab, map the incoming fields with the corresponding fields in the request structure.



   e. On the **Response Mapping** tab, map the incoming fields from the response structure with the output fields.



f. On the **Output Fields** tab, you can edit the precision and scale of the fields, if required.

g. On the **Advanced** tab, specify the cache size for the web service request and response.

5. To add a Target transformation, on the **Transformation palette**, click **Target**.

   a. On the **General** tab, enter a name and description for the target.

   b. Draw a link to connect the previous transformation to the Target transformation.

   c. Select the output group from the Web Services transformation that you want to connect to the target, and click **OK**.

   d. Click the **Target** tab, create a target at run time, specify the target object, and configure the advanced target properties.

   e. Click **Field Mapping** and view the fields that are automatically mapped to the target.

6. Click **Save and Run** to run the mapping.

   You can also configure a mapping task based on the mapping.

SAP BAPI and transformation data types

Use only the compatible SAP BAPI data types in mappings and mapping tasks to process SAP data through the BAPI function.

The following table lists the SAP BAPI data types that you can use with SAP BAPI Connector, along with the corresponding transformation data types:

SAP BAPI Data Type	Transformation Data Type	Range for Transformation Data Type
ACCP	Date/time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D.
CHAR	String	1 to 104,857,600 characters Fixed-length or varying-length string.
CLNT	String	1 to 104,857,600 characters Fixed-length or varying-length string.
CUKY	String	1 to 104,857,600 characters Fixed-length or varying-length string.

SAP BAPI Data Type	Transformation Data Type	Range for Transformation Data Type
CURR	Decimal	Precision 1 to 28 digits, scale 0 to 28
DATS	Date/time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision to the nanosecond.
DEC	Decimal	Precision 1 to 28 digits, scale 0 to 28
DF16_DEC	Decfloat16	Range of 1-15 and scaling of maximum 14. Decimal floating point number stored in BCD format.
DF34_DEC	Decfloat34	Range of 1-31 and scaling of maximum 30. Decimal floating point number stored in BCD format.
DF16_RAW	Double	Maximum of 16 positions with floating decimal. Decimal floating point number stored in binary format.
DF34_RAW	Double	Maximum of 34 positions with floating decimal. Decimal floating point number stored in binary format.
FLTP	Double	Precision 15, scale 0
INT1	Small Integer	Precision 5, scale 0
INT2	Small Integer	Precision 5, scale 0
INT4	Integer	Precision 10, scale 0
INT8	Int8	8-byte integer between -9,223,372,036,854,775,808 and +9,223,372,036,854,775,807. The length is set at 19 positions.
LANG	String	1 to 104,857,600 characters Fixed-length or varying-length string.
LCHR	String	1 to 104,857,600 characters Fixed-length or varying-length string.
NUMC	String	1 to 104,857,600 characters Fixed-length or varying-length string.
PREC	Binary	Uninterrupted sequence of bytes with a maximum length of 255 positions.
QUAN	Decimal	Precision 1 to 28 digits, scale 0 to 28
RAW	Binary	Uninterrupted sequence of bytes with a maximum length of 255 positions.
TIMS	Date/time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision to the nanosecond.
UNIT	String	1 to 104,857,600 characters Fixed-length or varying-length string.
VARC	String	1 to 104,857,600 characters Fixed-length or varying-length string.

Author

Anu Chandrasekharan

Acknowledgements

The author would like to acknowledge Sivaramakrishnan Kalyanaraman, for his technical assistance.